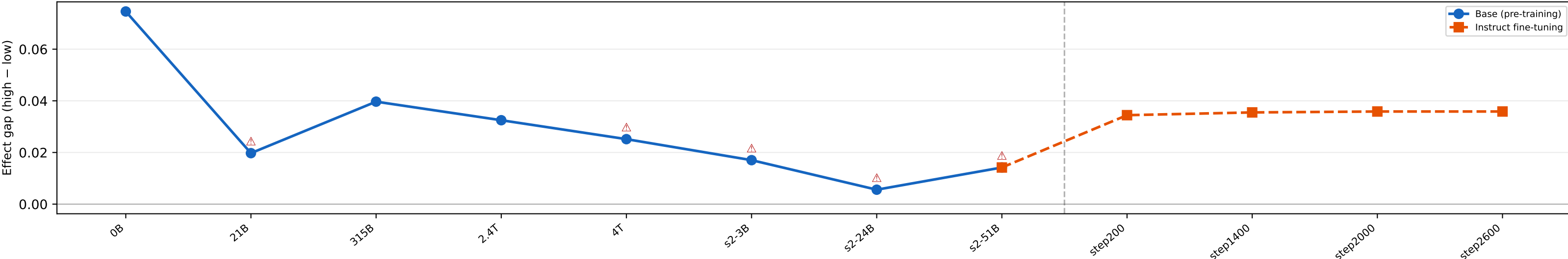
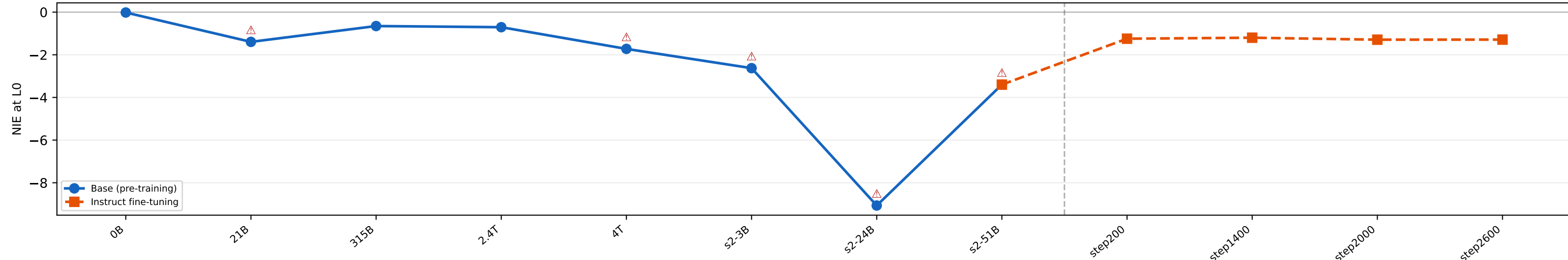


OLMo-2-0425-1B — Religion bias: learning trajectory
Left: base pre-training checkpoints | Right: instruction fine-tuning

Effect gap — how much corrupting subject tokens reduces bias-consistent probability
Larger = model relies more on subject identity for this domain



NIE at L0 — normalized indirect effect at the first transformer layer
= (Patched - Corrupted) / (Clean - Corrupted)



Raw causal signal at the first transformer layer — absolute scale
Reflects both L0 importance and overall bias strength

